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ORIGINAL RESEARCH

Large language models show promising performance for some systematic review tasks but call for cautious implementation: a systematic review

Florian Laignelot[a,1], Guillaume L. Martin[a,1], Mohamad Ossman[a], Ophélie Pingeon[a], Amine Boubaker[a], Emma Picovschi[a], Jia Kim[a], Xavier Tannier[b], Jérémie F. Cohen[c,d], Agnès Dechartres[a,*]

[a]*Sorbonne Université, INSERM, Institut Pierre Louis d'Epidémiologie et de Santé Publique, UMR-S 1136, AP-HP, Hôpital Pitié-Salpêtrière, Département de Santé Publique, Paris, France*

[b]*Sorbonne Université, Université Sorbonne Paris-Nord, Limics, Paris, France*

[c]*Centre for Research in Epidemiology and Statistics (Inserm UMR 1153), Université Paris Cité, Paris, France*

[d]*Department of General Paediatrics and Paediatric Infectious Diseases, Université Paris Cité, Hôpital Necker-Enfants Malades, APHP, Paris, France*

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Abstract

Objectives: With the exponential growth of biomedical literature, the challenge of conducting systematic reviews is becoming increasingly burdensome. We aimed to evaluate the performance of large language models (LLMs) in the automation of some or all steps of systematic reviews and meta-analyses.

Study Design and Setting: In this systematic review, we searched PubMed, Embase, the Cochrane Library and preprint platforms up to January 14, 2025. We included any studies assessing the performance of LLMs (eg, generative pre-trained transformer [GPT], Claude, Mistral) in any step of the systematic review process. Pairs of reviewers independently extracted data and assessed risk of bias. We conducted analyses using median (interquartile range [IQR]) for positive (PPA) and negative percent agreements (NPA), respectively, analogous to sensitivity and specificity, between LLMs and human reviewers.

Results: From 3889 unique references, we included 63 studies of which 52 reporting performance metrics for a total of 148 LLM performance assessments. Most assessments concerned GPT models (n = 114, 77%). The most frequently evaluated tasks were title and abstract screening (n = 78, 53%), data extraction (n = 23, 16%), and full-text screening (n = 20, 14%). For title and abstract screening, overall median PPA was 0.92 (IQR 0.69−0.98) and median NPA was 0.89 (0.72−0.95). For full-text screening, the overall median PPA was 0.93 (0.87−1.00) and median NPA was 0.92 (0.78−0.97). Late-generation LLMs released after GPT-4 seemed to provide higher performance than earlier models. For other tasks, authors reported overall good performances, but variability of performance metrics precluded complete quantitative synthesis. Global accuracy for data extraction tasks ranged from 0.36 to 1.00, with a median accuracy of 0.95 (IQR 0.91−0.97, n = 11). For the "risk of bias assessment" task, accuracy ranged from 0.44 to 0.90 (median = 0.62, IQR 0.53−0.76, n = 6).

Conclusions: The performance of LLMs, particularly newer generations, shows promise in automating some repetitive steps of systematic reviews such as screening. However, their successful integration will require appropriate safeguards and careful implementation. © 2026 The Author(s). Published by Elsevier Inc. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

Keywords: Artificial intelligence; Large language models; Systematic reviews; Meta-analyses; Methodology; Screening

1. Introduction

Systematic reviews and meta-analyses are considered among the most reliable forms of evidence when they provide comprehensive summaries of existing literature following strict methodological standards. Such studies require substantial human resources, time and skills [1]. In 2016, the estimated mean time to complete and

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[1] Contributed equally.

* Correspondance to: Sorbonne Université, INSERM, Institut Pierre Louis d'Epidémiologie et de Santé Publique, UMR-S 1136, Boulevard de l'Hôpital, Paris 75013, France.

E-mail address: agnes.dechartres@aphp.fr (A. Dechartres).

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Plain Language Summary

Systematic reviews are one of the most reliable ways to answer medical and public health questions. They bring together all available studies on a topic and help clinicians and policymakers make informed decisions. However, producing a high-quality systematic review takes a lot of time and effort. Whole teams of researchers spend months screening thousands of articles, extracting data, and double-checking results. With little more than a million of new publications every year, keeping reviews up to date is becoming increasingly difficult.

LLMs, such as ChatGPT, may help reduce this workload. These tools can read and summarize text and might assist with repetitive tasks like selecting relevant studies or extracting information from articles. But it is still unclear how reliable these tools are for research purposes. This is the first systematic review to assess LLMs' performance to facilitate systematic reviews. We sought to review all studies that tested LLMs in the different steps of systematic reviews and found 63 studies evaluating how well these tools performed compared with human reviewers. Overall, LLMs showed good agreement with humans for tasks such as screening titles and abstracts, and full-text articles. Newer models seemed to perform better than older ones. However, performance was more variable for complex tasks that require interpretation, such as extracting detailed data or assessing methodological quality. Our findings suggest that LLMs could help researchers work faster and make systematic reviews more efficient. However, they are not ready to replace human judgment. These tools can make mistakes, produce inconsistent results, or generate inaccurate information if not carefully supervised. In practice, LLMs should be used as assistants rather than substitutes. With proper safeguards, transparent reporting, and human oversight, they may become valuable tools to support evidence-based healthcare and help keep research up to date.

publish a systematic review was 67.3 weeks, with an average of five reviewers involved [2]. In light of the exponential growth of biomedical literature, the burden of conducting systematic reviews and keeping them up to date is expected to escalate quickly [3,4]. Today, with more than 1.2 million publications pouring into MEDLINE each year, reviewers are likely approaching the limits of human workload capacity, risking a point at which systematic reviews may no longer be ''systematically'' sustainable [5].

Innovative automation tools have naturally emerged to alleviate the reviewers' workload [6]. In 2022, Blaizot et al [7] reviewed the use of artificial intelligence (AI) tools in systematic reviews, and no algorithm appeared to be sufficiently reliable.

The latest advancements in generative AI, with the development of large language models (LLMs), might profoundly reshape the field. In 2022, OpenAI introduced generative pre-trained transformer 3 (GPT-3), an advanced LLM, which gained unprecedented attention owing to its capabilities and accessibility via an intuitive and free user interface, ChatGPT [8] and numerous other advanced models are now available. These tools are redefining processes across an increasing number of domains. Designed to provide contextualized answers while handling massive amounts of text, LLMs hold the potential to greatly alleviate reviewers' workload when screening thousands of references and extracting data from hundreds of articles. Recent publications have documented and analyzed the increasing use of LLMs in systematic reviews and experienced an LLM-assisted review, highlighting potential promises and limitations [9,10].

In this study, we evaluated the performance of LLMs in automating some or all steps of systematic reviews and meta-analyses in the field of biomedical research.

2. Methods

This systematic review followed the Cochrane Handbook and was reported in accordance with the PRISMA extension for diagnostic test accuracy (DTA) Studies [11]. The review protocol was registered in PROSPERO (CRD42024594795). Modifications are reported and justified in Appendix A.

2.1. Search strategy

A systematic literature search was conducted from inception to 14 January, 2025, on PubMed, Embase, and the Cochrane Library. Search terms included Medical Subject Headings terms (or Emtree terms for Embase) and free-text words, related to LLMs—such as but not limited to ''artificial intelligence,'' ''LLM,'' and ''ChatGPT''—and systematic reviews or meta-analyses such as ''systematic reviews as topic'' or ''meta-analyses.'' The search equation is in Appendix B. As an attempt to identify gray literature, we also screened preprint platforms such as the Medical Preprint Repository and Journal of Medical Internet Research preprints for unpublished studies. We did not systematically search preprints servers primarily dedicated to fundamental, preclinical, or technical sciences (eg, bioRxiv or arXiv) as our review focused on health and clinical settings in biomedical research.

What is new?

Key findings

- LLM performance show high overall agreement with human reviewers for screening tasks.
- Late-generation LLMs seemed to show better performance than earlier generations.

What this adds to what is known?

- This is the first systematic review assessing LLMs' performance to facilitate systematic reviews.
- LLMs show promising performance for screening tasks.

What is the implication and what should change now?

- LLMs could possibly facilitate the conduct of systematic reviews.
- Standardized performance metrics and reporting methods are needed, along with appropriate safeguards.
- Careful implementation of LLMs could be considered, associated with further research on less evaluated tasks such as Risk of Bias Assessment and data extraction.

2.2. Study selection

Pairs of reviewers (F.L./.O.P., A.B./E.P. or J.K.) independently screened titles and abstracts of all retrieved citations, then full-text reports using Rayyan [12]. Disagreements were resolved by discussion until consensus was reached with the assistance of a senior reviewer (A.D.).

We included studies assessing the performance of LLMs in automating any task of systematic reviews and meta-analyses in biomedical research. Definitions for included LLMs are available in Appendix C.

We systematically contacted authors of conference abstracts. Such abstracts were included only when extractable quantitative data were available or could be obtained from the authors. There were no restrictions on language, publication date, or type. We only did not consider systematic reviews, commentaries, or editorials.

2.3. Data extraction and quality assessment

Pairs of reviewers (F.L./M.O., O.P./A.B.) independently extracted data by using a standardized form (see Appendix C for the list of items) and assessed the risk of bias of the included quantitative studies. We did not assess the methodological quality of qualitative studies because of their variability. Because the quality assessment tool for AI-centered DTA studies, the QUADAS-AI tool, was under development [13], the closest framework we could use was the QUADAS-2 tool originally designed for DTA studies [14]. Following the Cochrane Handbook for Systematic Reviews of DTA studies [15], we tailored signaling questions to account for the specific characteristics of LLMs. Appendix D details and justifies all modifications made to the initial QUADAS.

2.4. Data synthesis

We performed descriptive analyses using frequencies and percentages for categorical variables and median (IQR) for continuous variables. Qualitative studies assessing LLM performance without quantitative metrics were summarized narratively. When quantitative studies reported multiple assessments such as the assessment of different LLMs (eg, GPT-3.5 and GPT-4), different datasets, or different review tasks, we considered each assessment separately. We defined head-to-head comparisons as assessments within the same study that reported the performance of two distinct LLMs or, for the same LLM, two distinct implementation methods, on the same dataset, for the same task, and against the same comparator.

We summarized the performance of LLMs using positive percent agreement (PPA) and negative percent agreement (NPA), which measure concordance between the LLM and human reviewers. We preferred PPA and NPA over sensitivity and specificity because these imply a comparison to a diagnostic reference standard, which is not the case for human reviewers. This approach is recommended by the Food and Drug Administration [16]. Specifically, PPA represents agreement on positive results, analogous to sensitivity, whereas NPA represents agreement on negative results, analogous to specificity. Whenever data were available, we recalculated PPA, NPA, and corresponding 95% confidence intervals from the contingency table. When contingency data were not available, reported PPA and NPA values were extracted and directly plotted. When a study performed multiple assessments for the same task, LLM, and implementation method on the same dataset, we selected the assessment yielding the highest PPA to avoid redundancy.

Because of substantial heterogeneity in performance metrics and evaluation settings, we decided not to report meta-analyses and to present results using medians (IQR) as recommended by the Cochrane Handbook for DTA studies [15]. We plotted individual assessment estimates in receiver operating characteristic (ROC) space (ie, PPA against 1−NPA) and overlaid an ROC-space curve to visually summarize the overall pattern of the trade-off between PPA and NPA, when feasible [15]. This curve was used as an illustrative aid only and should not be interpreted as a pooled meta-analytic summary estimate.

We performed subgroup analyses by LLM generation (''early'' defined as pre-GPT-4 models, vs ''late,'' defined

as GPT-4 and later models), implementation method (web-based user interface (Web-UI) vs application programming interface [API]), and risk of bias (high/unclear vs low). When the available data were sparse across several strata, the ROC-space curve was not plotted.

Lastly, we conducted several sensitivity analyses. First, we assessed whether our findings were affected by studies with multiple assessments. Using the original sample, we randomly resampled the data such that for each study reporting multiple assessments from different LLMs or

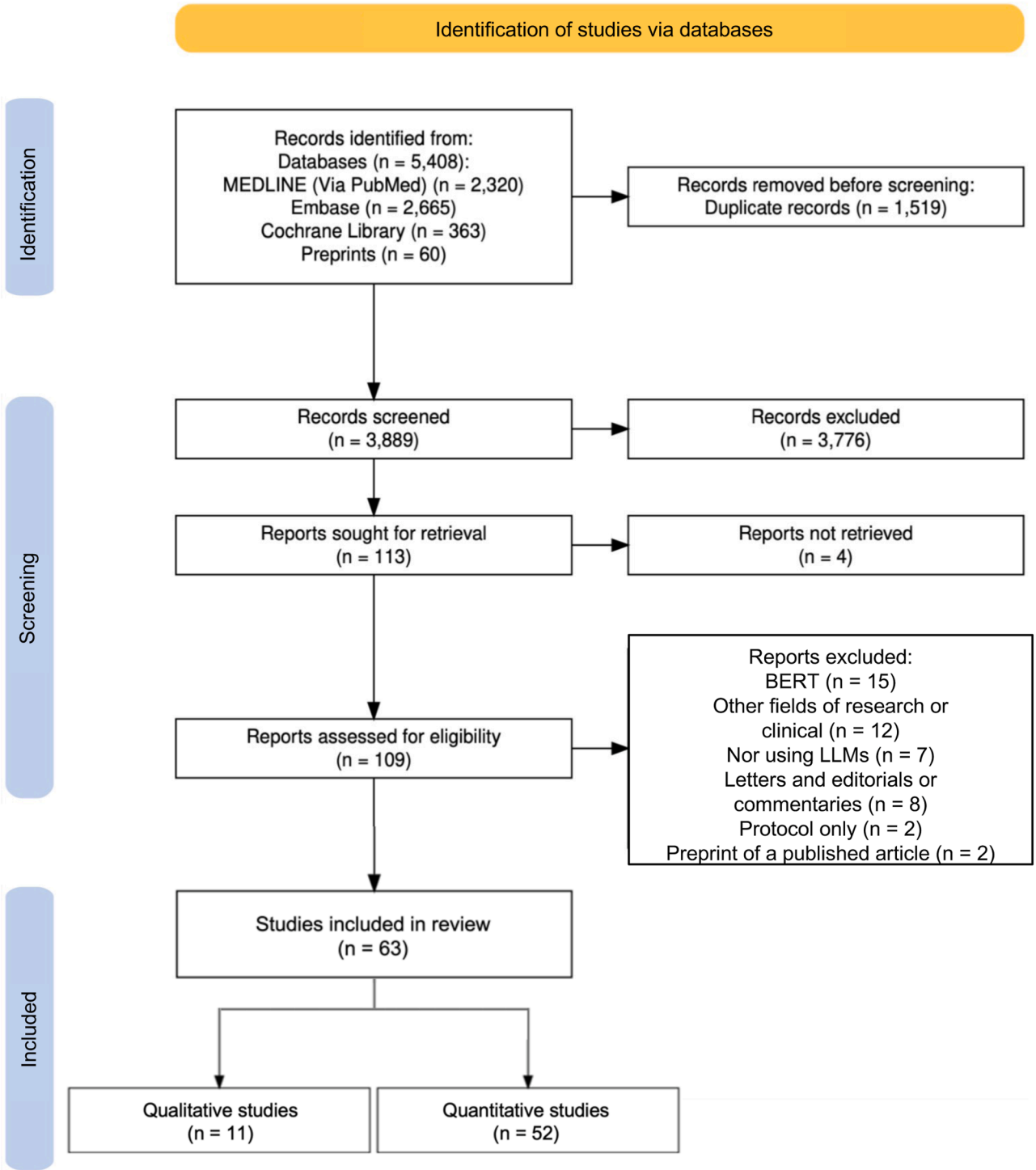


Figure 1. PRISMA Flow Diagram of study selection process.

Table 1. General characteristics of the included studies

Study characteristics	All studies ($N = 63$)	Qualitative studies ($N = 11$)	Quantitative studies ($N = 52$)
Year of publication			
2025	6 (10%)	0 (0%)	1 (2%)
2024	43 (68%)	4 (36%)	43 (83%)
2023	14 (22%)	7 (64%)	8 (15%)
Country of origin			
United States	14 (22%)	4 (36%)	10 (19%)
United Kingdom	12 (19%)	3 (27%)	9 (17%)
Canada	5 (8%)	0 (0%)	5 (10%)
China	3 (5%)	0 (0%)	3 (6%)
Japan	3 (5%)	0 (0%)	3 (6%)
Other	26 (41%)[a]	4 (36%)[b]	22 (42%)
Funding source			
Academic	21 (33%)	3 (27%)	18 (35%)
Private	2 (3%)	0 (0%)	2 (4%)
Both academic and private	3 (5%)	0 (0%)	3 (6%)
None	20 (32%)	3 (27%)	17 (33%)
Not reported	17 (27%)	5 (45%)	12 (23%)
First author affiliation			
Academic	57 (90%)	11 (100%)	46 (88%)
Private company	6 (10%)	0 (0%)	6 (10%)
Application field			
Multiple fields	20 (32%)	1 (9%)	19 (37%)
Oncology	5 (8%)	0 (0%)	5 (10%)
Neurology	4 (6%)	0 (0%)	4 (8%)
Plastic surgery	4 (6%)	3 (27%)	1 (2%)
Other[c]	26 (41%)	6 (55%)	20 (38%)
Not reported	3 (5%)	1 (9%)	2 (4%)

[a] One study each: Australia, Austria, Belgium, Brazil, Finland, Greece, The Netherlands, South Korea, Switzerland, Taiwan, Thailand, Turkey, United Arab Emirates, unreported. Two studies each: Denmark, France, Germany, India, Iran, Ireland.

[b] One study each: Belgium, Greece, India, Ireland.

[c] One study each: anesthesia, cardiology, emergency medicine, internal medicine, neuro-oncology, nutrition, reproductive health. Two studies each: Dermatology, Gastroenterology, Global Health, Infectious Diseases, Orthodontics, Orthopedics, Pharmacology, Psychiatry, Radiology, Urology.

dataset, only one was selected. Second, to assess a possible optimism bias related to our choice of the assessment yielding the highest PPA when a study performed multiple assessments for the same task, LLM, and implementation method on the same dataset, we selected one assessment at random. Third, we repeated the analyses after exclusion of studies published as preprints only as these studies may be of lower methodological quality.

All analyses were conducted in R (version 4.4.1) using the ggplot2, lme4, and robvis packages [17–20].

3. Results

3.1. Study selection and general characteristics

From a total of 5408 items identified, 3889 unique references were screened after duplicate removal with 63 studies finally selected, including six preprints and six conference abstracts (Fig 1). Most studies were published in 2024 ($n = 43$, 68%). Twenty-one (33%) studies were funded by academic sources and 20 (32%) reported no funding. To note, no studies were funded by LLM developers. Authors were mostly affiliated with academic departments ($n = 57$, 90%). One author in a single study was affiliated with an LLM developing company and held stocks in that company. Among the 63 included studies, 11 (18%) were qualitative and 52 were quantitative (Table 1). List of included studies is available in Appendix E.

3.2. Risk of bias of quantitative studies

Risk of bias was assessed as high in at least one domain for 13 studies (25%), unclear in at least one domain for 30 (57%), and low in all domains for 10 (19%). Domains most frequently rated at high risk of bias were ‘‘dataset selection’’ and ‘‘reference standard’’ in 7 (13%) studies each (Supplementary Fig 1).

Table 2. Characteristics of LLMs and comparisons in the included studies

Tasks, LLMs, and comparators	All studies (*N* = 63)[a]	Qualitative studies (*N* = 11)	Quantitative studies (*N* = 52)	Quantitative assessments (*N* = 148)
Automated task				
Research question	4 (6%)	2 (18%)	2 (4%)	2 (1%)
Provide a shortlist of references	4 (6%)	0 (0%)	4 (8%)	5 (3%)
Protocol drafting	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Search equation	5 (8%)	1 (9%)	4 (8%)	6 (4%)
Screen title and abstract	22 (37%)	0 (0%)	22 (42%)	78 (53%)
Screen full text	4 (6%)	0 (0%)	4 (8%)	20 (14%)
Data extraction	12 (19%)	1 (9%)	11 (21%)	23 (16%)
Risk of bias assessment	10 (16%)	1 (9%)	9 (17%)	13 (9%)
Create code for analysis	1 (2%)	1 (9%)	0 (0%)	0 (0%)
Article drafting	1 (2%)	0 (0%)	1 (2%)	1 (1%)
Performing a systematic review	5 (8%)	5 (45%)	0 (0%)	0 (0%)
Review update	0 (0%)	0 (0%)	0 (0%)	0 (0%)
LLM evaluated				
GPT	55 (87%)	9 (82%)	46 (88%)	114 (77%)
GPT-3.5	31 (49%)	6 (55%)	25 (48%)	39 (26%)
GPT-4	30 (48%)	2 (18%)	28 (54%)	72 (48%)
GPT, version not reported	4 (6%)	1 (9%)	3 (6%)	3 (2%)
Claude	6 (10%)	0 (0%)	6 (12%)	11 (7%)
Claude 2	3 (5%)	0 (0%)	3 (6%)	5 (3%)
Claude 3	3 (5%)	0 (0%)	3 (6%)	6 (4%)
LLaMA 2	4 (6%)	0 (0%)	4 (8%)	7 (5%)
Mistral 7B	3 (5%)	0 (0%)	3 (6%)	7 (5%)
Gemini	3 (5%)	1 (9%)	2 (4%)	3 (2%)
Other	5 (8%)[b]	1 (9%)[c]	4 (8%)	6 (4%)[d]
Implementation method				
Web-based UI	42 (67%)	11 (100%)	31 (59%)	50 (34%)
Application programming interface	18 (29%)	0 (0%)	18 (34%)	64 (43%)
Unreported	4 (6%)	0 (0%)	4 (8%)	34 (23%)
Technical design				
Prompt engineering	40 (63%)	1 (9%)	39 (75%)	126 (85%)
Prompting	20 (32%)	10 (91%)	10 (19%)	17 (11%)
Unreported	3 (5%)	0 (0%)	3 (6%)	5 (3%)
Comparator				
Human researchers	31 (49%)	/	31 (59%)	98 (66%)
Published systematic reviews	16 (25%)	/	16 (30%)	42 (28%)
Published Cochrane systematic reviews	6 (10%)	/	6 (12%)	8 (5%)
N/A[e]	11 (17%)	11 (100%)	/	/

GPT, generative pre-trained transformer; LLM, large language model; UI, user interface.

[a] Sums of percentages exceed 100% because some studies evaluated multiple tasks, LLM, automation methods, technical designs, or comparators.

[b] Two studies: Flan-T5. One study each: PaLM 2, Grok, Loon Lens.

[c] One study: Grok.

[d] Three studies: PaLM 2. Two studies: Flan-T5. One study: Loon Lens.

[e] Qualitative studies.

3.3. Assessment of performance of LLMs: methods

Of the 52 quantitative studies, 26 (50%) performed multiple assessments, for a total of 148 unique assessments (median of 3 [2–5] per study). Sixteen (31%) studies assessed several LLMs, 2 (4%) a single LLM to perform different tasks, 5 (10%) a single LLM on multiple datasets, and 2 (4%) assessed a single LLM with multiple prompt

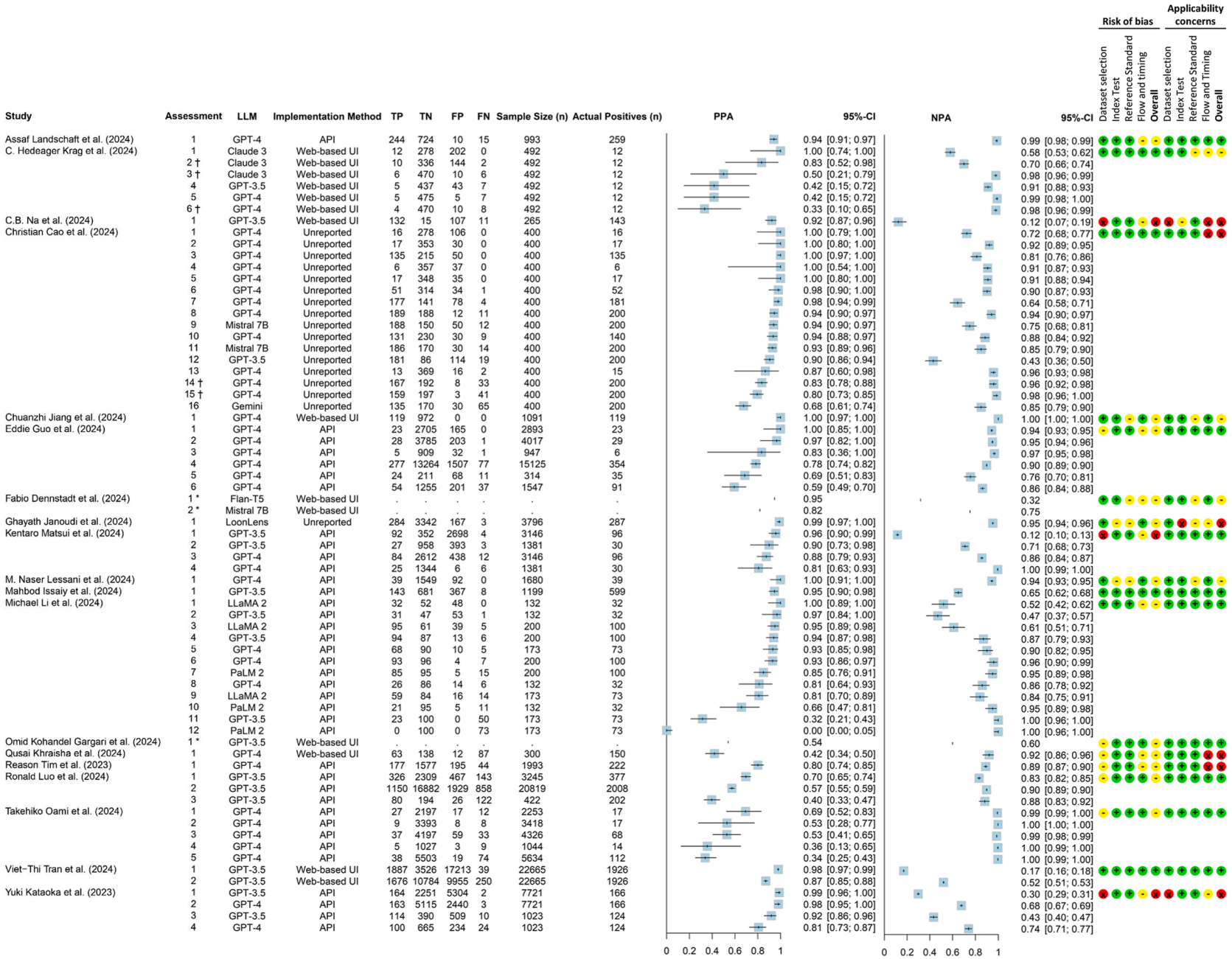


Figure 2. Abstract screening task coupled forest plot. LLM, large language model; API, application programming interface; UI, user interface; TP, true positive; TN, true negative; FP, false positive; FN, false negative; PPA, positive percent agreement; NPA, negative percent interval.

Table 3. LLM performances for title and abstract screening

		Stratified by LLM generation		Stratified by implementation method		Stratified by risk of bias	
Metric	Global	Early gen LLM	Late gen LLM	API	Web-based UI (or unreported)	High risk or unclear	Low risk
PPA median (IQR)	0.92 (0.69, 0.98)	0.91 (0.66, 0.95)	0.93 (0.69, 0.98)	0.82 (0.66, 0.95)	0.94 (0.89, 1.00)	0.83 (0.66, 0.95)	0.95 (0.90, 1.00)
NPA median (IQR)	0.89 (0.72, 0.95)	0.58 (0.40, 0.87)	0.91 (0.85, 0.96)	0.89 (0.74, 0.96)	0.88 (0.68, 0.92)	0.90 (0.76, 0.97)	0.85 (0.65, 0.91)

API, application programming interface; PPA, positive percent agreement; NPA, negative percent agreement; LLM, large language model; UI, user interface.

engineering strategies (ie, prompting strategies to maximize sensitivity/PPA or specificity/NPA). One additional study assessed several LLMs for different tasks, for a total of 31 assessments. Details on assessments performed are described below, in Table 2 and in Supplementary Table 1.

3.3.1. Systematic review task automation

The most common tasks for which LLMs were evaluated were title and abstract screening in 78 (53%) assessments, data extraction in 23 (16%), full-text screening in 20 (14%), and risk of bias assessment in 13 (9%).

3.3.2. Large language models

GPT was the most evaluated LLM, with 114 (77%) assessments. Overall, early-generation LLM were used in 44 assessments (30%) and late-generation LLM in 102 (70%).

3.3.3. Comparator details

LLMs were predominantly compared to human researchers performing identical tasks (n = 98, 66%). Other assessments compared LLM outputs with those from already published systematic reviews (n = 42, 28%), and eight assessments (5%) focused on Cochrane reviews only.

3.3.4. Reporting of performance metrics and contingency data

Authors reported a variety of performance metrics such as accuracy, F1-Score, sensitivity, specificity, and raw agreement, but did not systematically report contingency data.

3.4. Assessment of performance of LLMs: results

3.4.1. Screening titles and abstracts

From the 78 distinct assessments of the performance of LLMs in screening titles and abstracts, 69 provided PPA and NPA measures (Fig 2). The overall median PPA was 0.92 (IQR 0.69−0.98) and median NPA was 0.89 (0.72−0.95) (Table 3, Fig 3). Late-generation LLMs had higher NPA than early-generation models (0.91[0.85−0.96] vs 0.58 [0.40−0.87]), while PPA was similar between generations. Performance also varied by implementation method, with higher PPA observed for web-based interfaces than APIs (0.94[0.89−1.00] vs 0.82 [0.66−0.95]). Results were comparable across risk-of-bias groups. In five head-to-head comparisons, late-generation LLMs consistently had higher PPA and NPA than early-generation models (Supplementary Fig 2).

3.4.2. Screening full text

All 20 assessments of LLM performance in full-text screening provided PPA and NPA measures (Fig 4). The overall median PPA was 0.93 (0.87−1.00) and median NPA was 0.92 (0.78−0.97) (Table 4, Fig 5). Late-generation models again outperformed early-generation models, particularly for NPA (0.95 [0.90−0.98] vs 0.54 [0.48−0.64]). Web-based UI showed higher performance than APIs for both PPA (0.94 [0.92−1.00] vs 0.79 [0.71−0.87]) and NPA (0.95 [0.84−0.98] vs 0.54 [0.48−0.68]).

3.4.3. Other tasks

For other tasks, authors reported overall good performance. Accuracy for identifying novel research questions ranged from 0.68 to 0.83 (median = 0.74, IQR 0.71−0.79, n = 3). For the ‘‘search equation’’ task (n = 2), accuracy was 0.78 and 0.89.

Performance for the ‘‘data extraction’’ task assessments showed greater variability, with global accuracy ranging from 0.36 to 1.00, and a median accuracy of 0.95 (IQR 0.91−0.97, n = 11).

For the ‘‘risk of bias assessment’’ task, accuracy ranged from 0.44 to 0.90 (median = 0.62, IQR 0.53−0.76, n = 6).

Sensitivity analyses found consistent findings (data not shown).

An overview of qualitative studies is available in Appendix F.

4. Discussion

To our knowledge, this is the first systematic review assessing the potential interest of LLMs in facilitating the conduct of systematic reviews. Our results show that LLMs hold potential for automating some key steps of systematic

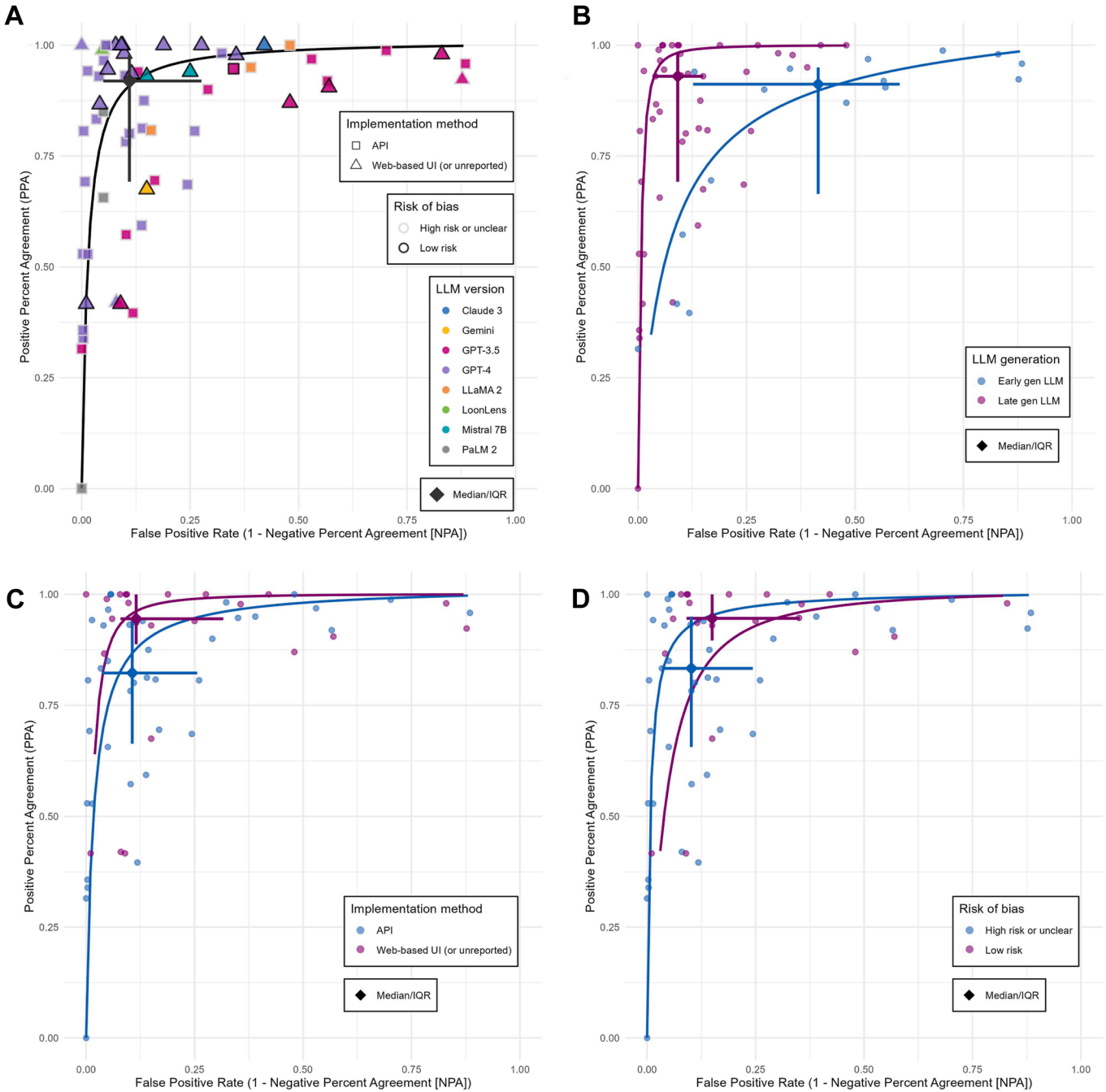


Figure 3. Receiver operating characteristic (ROC) plots of LLM performance for title and abstract screening. Each point represents a study-level performance estimate, with the y-axis showing positive percent agreement (PPA) and the x-axis the false positive rate (1—negative percent agreement, NPA). Solid curves represent ROC-space curves for descriptive purposes to visually summarize the overall pattern of the trade-off between PPA and NPA. This curve was used as an illustrative aid only and was not interpreted as a pooled meta-analytic summary estimate. Subpanels display overall results (A), and results stratified by LLM generation (B), risk of bias (C), and implementation method (API vs web-based interface) (D). LLM, large language model; API, application programming interface; UI, user interface.

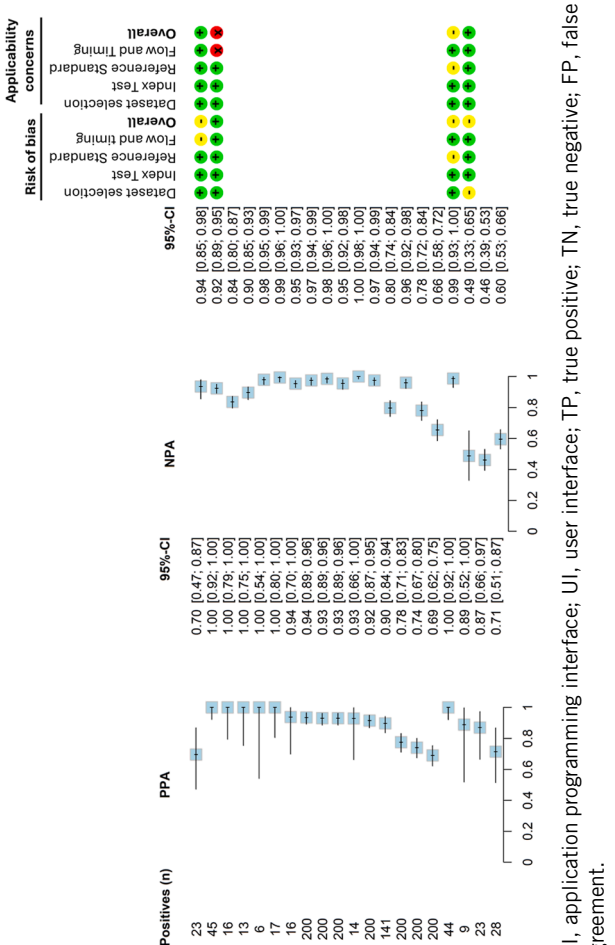


Figure 4. Full-text screening task coupled forest plot. LLM, large language model; API, application programming interface; UI, user interface; TP, true positive; TN, true negative; FP, false positive; FN, false negative; PPA, positive percent agreement; NPA, negative percent agreement.

reviews. LLMs, especially GPT were most frequently evaluated for title and abstract or full-text screening with results showing high overall agreement with human reviewers. Late-generation LLMs seem to outperform earlier generations, particularly for NPA, suggesting significant advancements in their capacity to correctly identify irrelevant information.

As compared with machine or deep learning tools, LLMs show markedly improved performance for screening [7]. The agreement between LLMs and human reviewers we observed is comparable or even better than inter-human agreement, which is frequently moderate. For title and abstract screening, single reviewers could fail to identify up to 13% of eligible studies [21] with moderate agreement between reviewers [22,23]. This suggests that LLMs may operate within the bounds of expected human variability for screening. In its latest version, the Cochrane Handbook already acknowledged the potential for LLMs to automate study selection and other parts of the review process, highlighting the need for assessment of these tools [24]. LLMs, particularly late-generation models, achieved performance levels suggesting they could feasibly substitute for one of the human reviewers for screening.

LLMs seem less performant for data extraction or risk of bias assessment with significant performance variability. This can be explained by the complexity of such tasks for LLMs, which necessitate interpretation and inference rather than information retrieval alone. Human reproducibility is also lower for these tasks. In the data extraction process, human error rates as high as 50% have been reported, and agreement between reviewers at the meta-analysis level could be as low as 31%, potentially affecting effect estimates [25,26]. Similarly, risk of bias assessments using the Cochrane RoB show low-to-fair reliability between human reviewers [27−31].

Our results suggest that implementation approach may influence LLM performance, with better results observed for web-based UIs. APIs reduce user-variability offering reproducibility and automation but execute only the instructions provided, whereas web-based UIs incorporate preoptimized prompting, guided workflows, and contextual memory, which may support more accurate outputs. Of note, risk of bias was higher for studies reporting use of APIs than other studies. Given the technical complexity of APIs, these investigations may have been predominantly conducted by data scientists or engineers, who might be less familiar with methods to minimize bias.

Interestingly, we found no studies using LLM models or tools specifically designed for systematic reviews. Several dedicated tools, such as Elicit or Systematiq AI, are now available but no study has evaluated their performance. Such purpose-built solutions may help standardize LLM implementation in the workflow of systematic reviews in future applications, but they need to be rigorously evaluated.

Table 4. LLM performances for full-text screening

		Stratified by LLM generation		Stratified by implementation method		Stratified by risk of bias	
Metric	Global	Early-generation LLM	Late-generation LLM	API	Web-based UI (or unreported)	High risk or unclear	Low risk
PPA median (IQR)	0.93 (0.87, 1.00)	0.80 (0.73, 0.87)	0.94 (0.92, 1.00)	0.79 (0.71, 0.87)	0.94 (0.92, 1.00)	0.87 (0.71, 0.89)	0.94 (0.91, 1.00)
NPA median (IQR)	0.92 (0.78, 0.97)	0.54 (0.48, 0.64)	0.95 (0.90, 0.98)	0.54 (0.48, 0.68)	0.95 (0.84, 0.98)	0.60 (0.49, 0.94)	0.94 (0.83, 0.98)

API, application programming interface; PPA, positive percent agreement; NPA, negative percent agreement; LLM, large language model; UI, user interface.

Our results are consistent with existing reviews reporting increasing research on LLMs in systematic reviews and proofs-of-concepts of LLM-assisted reviews [9,10]. However, although LLMs hold great promise, important points of attention remain.

First, despite the number of assessments identified, variability in methods and reporting across studies made a synthesis challenging. Greater adherence to emerging reporting frameworks for AI in health research, such as CONSORT-AI or STARD-AI, may improve transparency, reproducibility, and future evidence synthesis by encouraging clearer descriptions of datasets, model development, evaluation procedures, and sources of bias [32,33].

Second, with the lack of shared evaluation datasets, comparing LLM performance across studies and over time is difficult. The development of openly available, well-annotated benchmarking datasets, covering core tasks such as title/abstract and full-text screening and data extraction, is essential.

Third, reproducibility in LLM-oriented research is a fundamental concern. Unlike traditional algorithms, LLMs are not deterministic; the same prompt may yield different outputs depending on the underlying model architecture, session context, or even day-to-day backend updates. Of note, technical specifications such as model versions and prompt details were scarcely reported. Considering intense variability between model sizes and number of parameters, future research should include more details about LLMs configurations and investigate the reproducibility of LLM responses across updated versions, different implementations, or deployment settings.

Fourth, hallucinations also remain a key limitation of LLMs. Although such errors are reflected in quantitative performance metrics, their qualitative impact may not be fully captured by accuracy measures alone. Few studies have investigated why LLMs make errors, misclassify, whether performance varies by clinical domain, or how sensitive they are to ambiguous information. Although recent approaches such as retrieval-augmented generation and source-grounded prompting aim to reduce hallucinations by constraining outputs to verifiable information, these safeguards remain imperfect [34] and were not reported in included studies.

All these points need to be addressed to limit a ''black box'' effect and increase the confidence we could have in LLMs. LLMs should not be used because they are novel but because they are validated in this context. One major risk lies in their potential misuse to rapidly generate systematic reviews and meta-analyses. Many systematic reviews and meta-analyses already feature poor methodological quality, which could be exacerbated with the uncritical use of LLMs, without a solid understanding of both systematic review methods and the capabilities and pitfalls of LLMs. Therefore, the integration of LLMs should be accompanied by clear guidelines, rigorous validation processes, training of researchers, checklists, and periodic reassessments of their performance to ensure the integrity and quality of the systematic reviews produced.

Our study has limitations. First, we faced challenges due to variability in performance metrics across studies, limiting the ability to conduct meta-analyses. Optimism bias may have been introduced by our choice of the assessment yielding to the highest PPA when there were repeated assessments with some reporting near-perfect performance. Sensitivity analyses yielded similar results, suggesting this did not drive the findings.

Reporting bias cannot be excluded, which may have led to overly favorable estimates. Also, our choice to include qualitative studies made a synthesis challenging but we aimed to ensure a comprehensive overview of all studies on this topic. Finally, we acknowledge the time gap since our search as a limitation in a fast-evolving domain. However, research on LLM show a general trend of increasing capability over time, with newer models usually achieving higher performance than earlier generations [35]. Furthermore, users get more efficient with LLMs as they familiarize with them, and user-friendly tools emerge [36]. These trends support using our review as a baseline reference for LLM performances in systematic reviews, providing context for future studies as models continue to evolve.

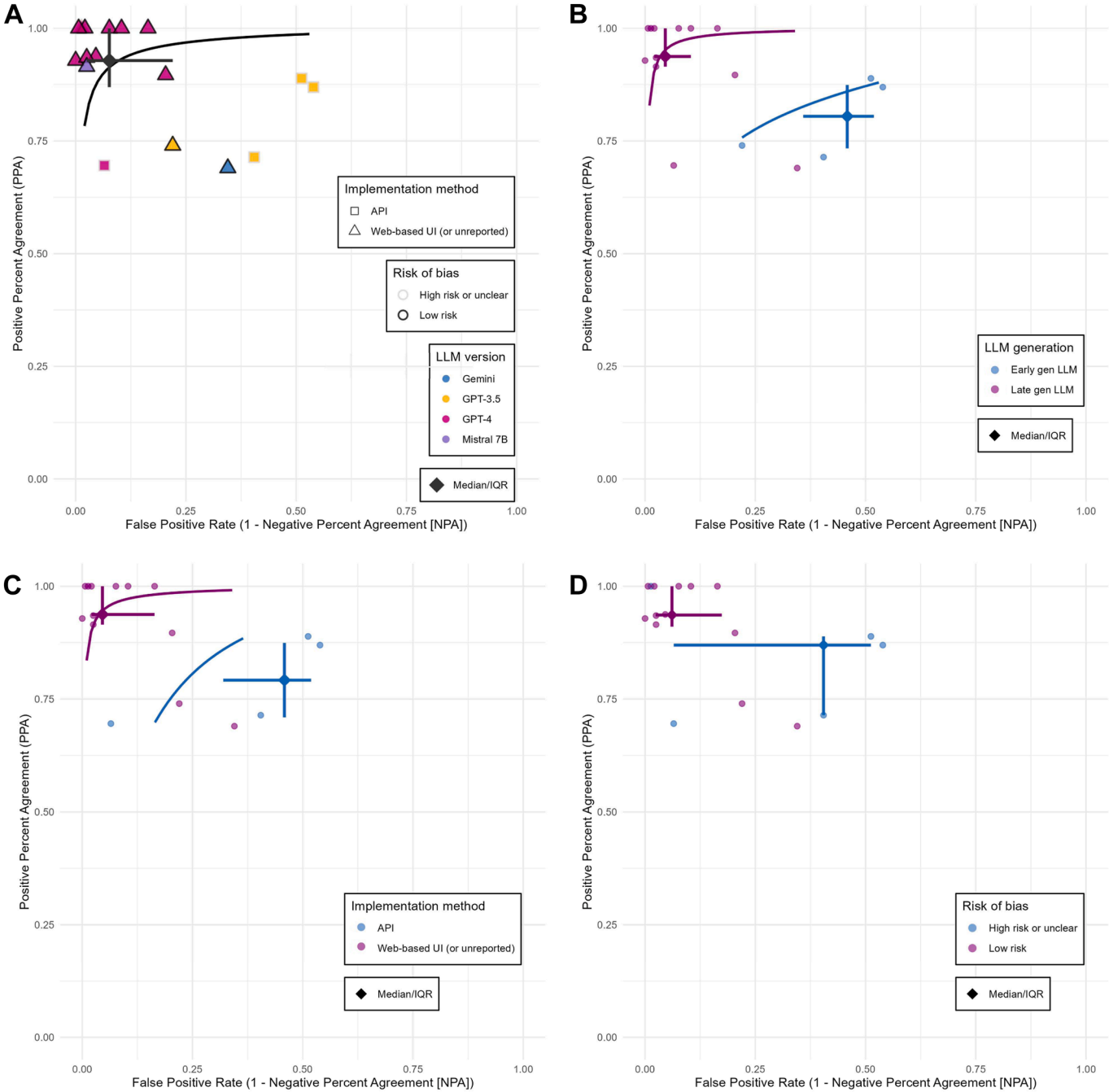


Figure 5. Receiver operating characteristics (ROC) plots of LLM performance for full-text screening. Each point represents a study-level performance estimate, with the y-axis showing positive percent agreement (PPA) and the x-axis the false positive rate (1—negative percent agreement, NPA). Solid curves represent ROC-space curves for descriptive purposes to visually summarize the overall pattern of the trade-off between PPA and NPA. This curve was used as an illustrative aid only and was not interpreted as a pooled meta-analytic summary estimate. Subpanels present overall results (A), and results stratified by LLM generation (B), and implementation method (API vs web-based interface) (C). LLM, large language model; API, application programming interface; UI, user interface.

5. Conclusion

The performance of LLMs, particularly newer generations, shows promise for screening in systematic reviews. However, their successful integration will require appropriate safeguards, standardization, and careful implementation.

Ethics approval

Not applicable, as this is based on a systematic review and analysis of aggregated data.

CRediT authorship contribution statement

Florian Laignelot: Writing − original draft, Investigation, Data curation, Conceptualization. **Guillaume L. Martin:** Writing − original draft, Methodology, Formal analysis, Conceptualization. **Mohamad Ossman:** Writing − review & editing, Investigation, Conceptualization. **Ophélie Pingeon:** Writing − review & editing, Investigation. **Amine Boubaker:** Writing − review & editing, Investigation. **Emma Picovschi:** Writing − review & editing, Investigation. **Jia Kim:** Writing − review & editing, Investigation. **Xavier Tannier:** Writing − review & editing, Validation, Conceptualization. **Jérémie F. Cohen:** Writing − review & editing, Validation, Methodology, Conceptualization. **Agnès Dechartres:** Writing − original draft, Validation, Supervision, Methodology, Conceptualization.

Declaration of competing interest

Guillaume Martin is currently an employee of Synapse Medicine, a French company that develops clinical decision-support medical devices integrated in electronic health records, as well as MedGPT.fr, an AI system designed to retrieve and synthesize medical guidelines to assist healthcare professionals in evidence-based decision-making. The other authors declare no competing interest.

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Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.jclinepi.2026.112221.

Data availability

Data will be made available on request.

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